



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

AUTOCAD BASIC SKILLS

LSVBA302

Apply AutoCAD basic skills

Competence

RQF Level: 3

Learning Hours

Credits: 6



Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV3001- TVET Certificate 3 in Land Surveying

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Purpose statement	This module is designed for students pursuing a level 3 TVET certificate in Land Surveying. It describes the skills, knowledge, and attitudes required to use AutoCAD Software.					
	At the end of this module, the students will be able to install the AutoCAD software student version, use AutoCAD features and function keys, and produce AutoCAD 2D drawings.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language; Linear distance measurement; Construction drawings interpretation; Building setting out; Angular measurements.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		20%	Formative assessment	20%	50%
	Practical work:		80%		80%	
	• Group project and presentation	30%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria	
Elements of competency	Performance criteria
1. Describe and install AutoCAD software	1.1. AutoCAD software is properly described according to Autodesk.
	1.2. AutoCAD software is successfully installed on the computer according to the type of student version.
	1.3. User interface is properly identified, navigated through, and customized in accordance with the task to be performed.
2. Use AutoCAD features and function keys	2.1. Toolbars are properly used based on the given task.
	2.2. AutoCAD features are properly identified and used according to the assignment to be performed.
	2.3. AutoCAD function keys and their shortcuts are appropriately utilized.

3. Produce AutoCAD 2D drawings	3.1. Settings are properly selected according to the drawing requirements.
	3.2. 2D drawings are effectively designed according to the given assignment.
	3.3. Drawing Layouts are appropriately plotted and printed according to the designed drawings.

Course content

Learning outcomes	At the end of the module the learner will be able to:
	1. Describe and install AutoCAD software 2. Use AutoCAD features and function keys 3. Produce AutoCAD 2D drawings

Learning outcome 1: Describe and install AutoCAD software	Learning hours: 7
Indicative content	
* Description of AutoCAD software (<i>Refer to ISO 128-all parts</i>) • AutoCAD software • Usage of AutoCAD (2D 2d models) • Application of AutoCAD * Installation of AutoCAD student version • Identify the computer setup system (32 or 64 bit) • Identify AutoCAD versions (current version preferred if computer specifications allow) • Register or create an trainee account at :https://www.autodesk.com/education/edu-software/ • Select the AutoCAD version and software requirements matching with computer specifications • Download the AutoCAD free student version setup based on the computer operating system • Install AutoCAD software in the computer following instructions provided by Autodesk * Navigation and customization of the AutoCAD user interface • Identify the user interface elements (workspace, menu bar, toolbars, status bar,	

commanding area) • Customize the user interface (CUI) to improve accessibility and reduce the number of steps for frequent tasks.	
Resources required for the learning outcome	
Equipment	• Computer, computer lab, projector
Materials	• Marker pen, pieces of chalk, pieces of paper, duster, whiteboard, flip chart
Tools	• AutoCAD software, storage device
Facilitation techniques	• Trainer guided • Individual and group work • Practical exercise
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 2: Use AutoCAD features and function keys		Learning hours: 25
Indicative content		
* Use AutoCAD toolbars • Draw, dimension, layer, modify, format * Use AutoCAD features • Menu bar features, annotation/labeling, layout * Use AutoCAD shortcuts and function keys • Manage screen (e.g.: tool palette, property palette, clean screen,) • Manage drawing (e.g.: new, open, save the drawings, plot dialog box,) • Manage workflow (e.g.: copy, cut, paste, undo, redo) • General feature (e.g.: pick a style, toggle cords, toggle grid) • Toggle drawing mode (e.g.: functions keys: toggle grid mode, ortho mode, snap mode, text screen)		
Resources required for the indicative content		
Equipment	Computer, computer Lab, projector	

Materials	Marker pen, pieces of chalk, pieces of paper, duster/white-board, flip chart etc...
Tools	AutoCAD software, storage device
Facilitation techniques	• Trainer guided • Individual and group work • Practical exercise
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 3: Produce AutoCAD 2D Drawings		Learning hours: 28
Indicative content		
* Settings preferences • Working units (metric, imperial, ...) • Layer • Scale • Direction * Design 2D drawings • Geometric figures and objects • Lines and curves • Plan views (layout plan, site plan) • Section view • Side view * Plotting and printing drawing layout • Create and update drawing layout (Title, Legend, Title block, Grid, Orientation) • Plotting and printing layout drawing		
Resources required for the indicative content		
Equipment	Computer, Computer lab, projector	
Materials	Marker pen, pieces of chalk, pieces paper, duster, whiteboard, flip chart etc...	
Tools	AutoCAD software, storage device	

Facilitation techniques	• Trainer guided • Individual and group work • Practical exercise
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Integrated/Summative Assessment

Integrated situation

RHINO CONSTRUCTION Ltd company is contracted to produce architectural drawings of a single story residential building accommodating one family in Musanze District. The contract scope is to produce drawings of the building. The building comprises three (3) bedrooms, one (1) living room, one (1) dinning room and two (2) bathrooms. As a designer and employee of the company, you are requested to reproduce the site plan and floor plan as given in hard copy using AutoCAD and produce drawing in DWG and PDF format. This work is supposed to be done in 5 hours. Consider the following information:

- * Metric system is used for dimensioning
- * Two car parking of 7m by 6m is on the front side of the building;
- * The plot size is 25m by 20m.
- * The drawing scale is 1:100

Resources

Equipment	Computer, computer lab, projector, printer
Materials/ Consumables	Markers, pieces of chalk, pieces of paper, a duster
Tools	AutoCAD software, storage device, marker/chalkboard

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: (10%)	The AutoCAD interface is customized.	The menu bar is customized			2
		Draw toolbar is customized			2
		Modify toolbar is customized			2
		Dimension toolbar is customized			2

		Layer toolbars are customized on the interface			2
Learning outcome 2: (30%)	The AutoCAD features are used.	Line and/or polyline are used			4
		Modify toolbar is used			7
		Dimension feature is used			4
		Hatch feature is used			3
		Text feature is used			3
		Block feature is used			2
		Layer feature is used			7
Learning outcome 3: (60%)	The floor plan is reproduced	The metric unit is used			3
		A scale of 1:100 is respected			3
		The legend is shown			4
		The floor plan is well dimensioned			4
		The labelling is created and readable			4
		The title block is created			4
		The document is saved in DWG and PDF format			5
		Orientation, Title and Grid are well organized			3
	The site plan is reproduced	The metric unit is used			3
		The scale of 1:100 is respected			3
		The legend is shown			4
		The site plan is well dimensioned			4
		The labeling is created and readable			4
		The title block is created			4
		The document is saved in DWG and PDF format			5
		Orientation, Title and Grid are well organized			3
Total marks		100			
Percentage Weightage		100%			

References:

1. Alain J.KALAMEJA. (2009) “AutoCAD 2010 Tutor for Engineering Graphics”.
2. Gopalakrishna, K.R (1998). Engineering Drawing. Delhi: Subash Publication.
3. Gupta, R.G. & Prakashan, S. (1995), A textbook of Engineering Drawing. Delhi: Khana Publisher
4. Venugopal, K. (2004). Engineering drawing and graphics + AutoCAD. New Delhi: New Age
5. Wilson, R. (1990). Engineering Drawing and Design. New York: Edward Arnold
6. ISO 128-All parts technical product documentation (TPD) — general principles of representation — part 1: introduction and fundamental requirements.